In [2]:
```
!pip install seaborn
```

```
Collecting seaborn
  Downloading seaborn-0.13.2-py3-none-any.whl.metadata (5.4 kB)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in /opt/miniconda3/lib/p
ython3.14/site-packages (from seaborn) (2.5.2)
Requirement already satisfied: pandas>=1.2 in /opt/miniconda3/lib/python3.1
4/site-packages (from seaborn) (3.0.5)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in /opt/miniconda3/li
b/python3.14/site-packages (from seaborn) (3.11.1)
Requirement already satisfied: contourpy>=1.0.1 in /opt/miniconda3/lib/pytho
n3.14/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /opt/miniconda3/lib/python3.1
4/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in /opt/miniconda3/lib/pyth
on3.14/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /opt/miniconda3/lib/pyth
on3.14/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /opt/miniconda3/lib/python
3.14/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (26.0)
Requirement already satisfied: pillow>=9 in /opt/miniconda3/lib/python3.14/s
ite-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (12.3.0)
Requirement already satisfied: pyparsing>=3 in /opt/miniconda3/lib/python3.1
4/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /opt/miniconda3/lib/p
ython3.14/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post
0)
Requirement already satisfied: six>=1.5 in /opt/miniconda3/lib/python3.14/si
te-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn)
(1.17.0)
Downloading seaborn-0.13.2-py3-none-any.whl (294 kB)
Installing collected packages: seaborn
Successfully installed seaborn-0.13.2
```

In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [3]:
```
df = pd.read_excel('sales_data (2).xlsx')

print(df.head())
print(df.isnull().sum())
df['Sales'] = df['Sales'].fillna(df['Sales'].mean())
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
print(df.describe())
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()

print(product_summary)
```

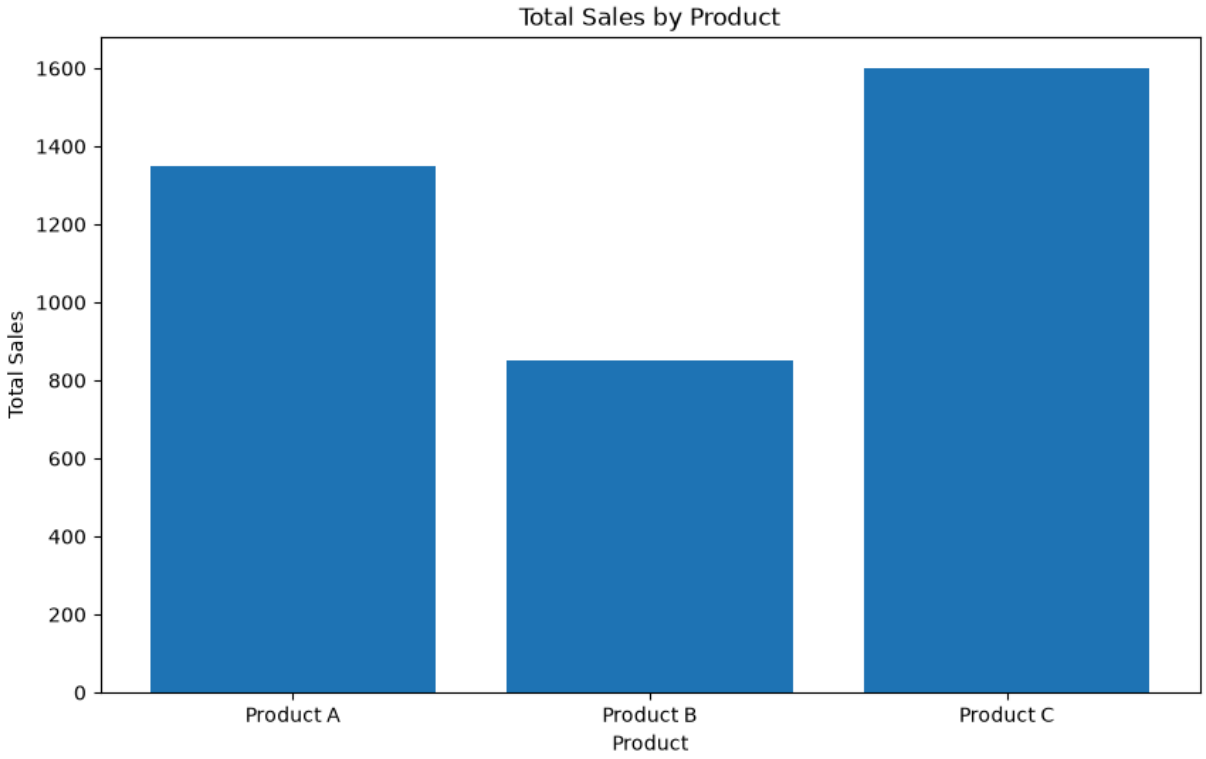
```
                  Date    Product  Sales  Quantity Region
0  2023-01-01 00:00:00  Product A    200         4  North
1  2023-02-01 00:00:00  Product B    150         3  South
2  2023-03-01 00:00:00  Product A    220         5  North
3  2023-04-01 00:00:00  Product C    300         6   East
4  2023-05-01 00:00:00  Product B    180         4   West
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
            Sales   Quantity
count   16.000000  16.000000
mean   237.500000   5.375000
std     64.031242   1.746425
min    150.000000   3.000000
25%    187.500000   4.000000
50%    225.000000   5.500000
75%    302.500000   7.000000
max    340.000000   8.000000
     Product  Sales  Quantity
0  Product A   1350        33
1  Product B    850        17
2  Product C   1600        36
```

In [13]:
```
plt.figure(figsize=(10, 6))

plt.bar(product_summary['Product'], product_summary['Sales'])

plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')

plt.show()
```



In [14]:
```
df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)

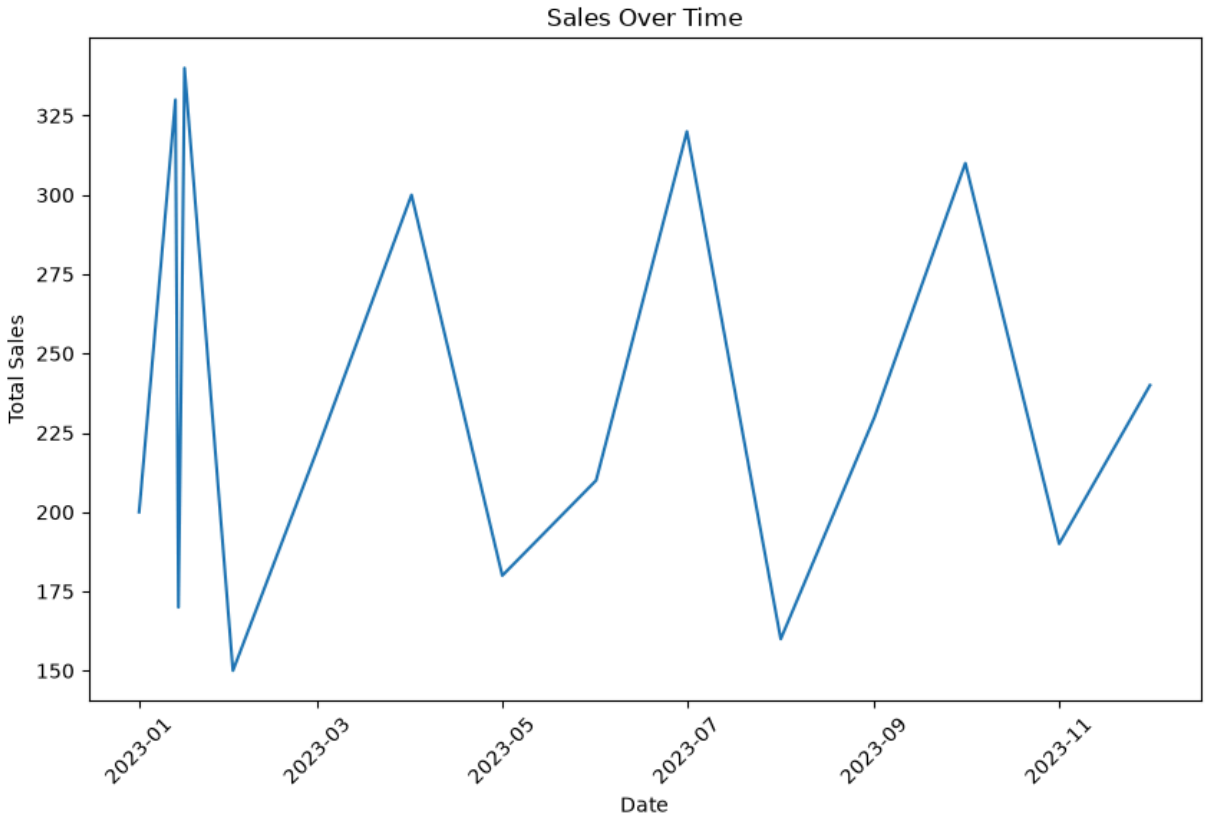
sales_over_time = df.groupby('Date')['Sales'].sum().reset_index()

plt.figure(figsize=(10, 6))

plt.plot(sales_over_time['Date'], sales_over_time['Sales'])

plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')

plt.xticks(rotation=45)
plt.show()
```



In [15]:
```
pivot_table = df.pivot_table(
    values='Sales',
    index='Region',
    columns='Product',
    aggfunc='sum',
    fill_value=0
)

print(pivot_table)
```

```
Product  Product A  Product B  Product C
Region
East             0          0       1600
North         1350          0          0
South            0        480          0
West             0        370          0
```

In [16]:
```
correlation_matrix = df.corr(numeric_only=True)

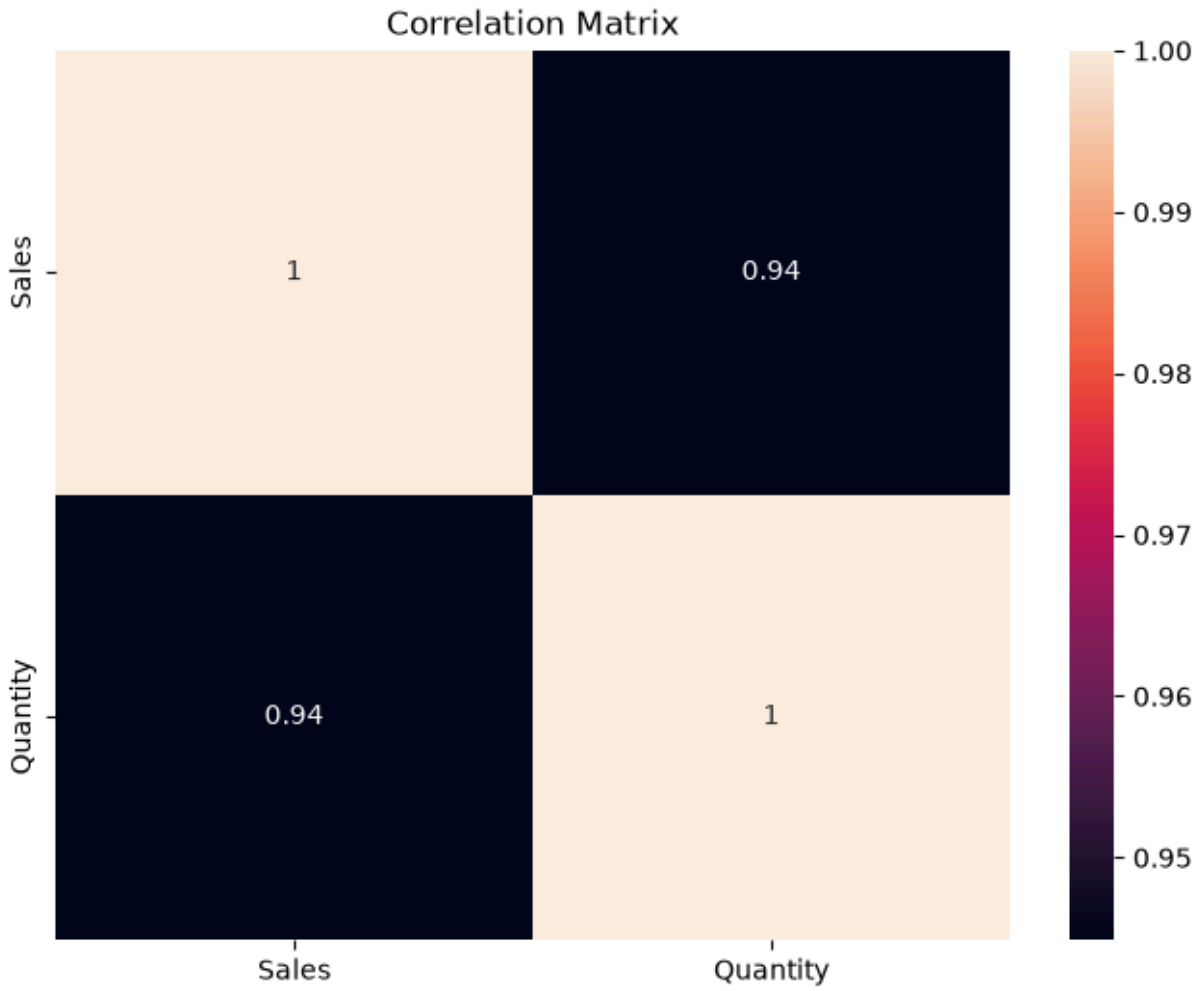
print(correlation_matrix)

plt.figure(figsize=(8, 6))

sns.heatmap(correlation_matrix, annot=True)

plt.title('Correlation Matrix')
plt.show()
```

```
              Sales  Quantity
Sales      1.000000  0.944922
Quantity   0.944922  1.000000
```


Correlation Matrix
Sales
Quantity
1
0.94
0.94
1
Sales
Quantity
1.00
0.99
0.98
0.97
0.96
0.95

In [ ]: